INCH-POUND
MIL-PRF-49450B(CR)
10 March 2000
SUPERSEDING
MIL-PRF-49450A(CR)
9 October 1997

PERFORMANCE SPECIFICATION

BATTERY, RECHARGEABLE, NICKEL-CADMIUM, VENTED, AIRCRAFT.

This specification is approved for use by the USACECOM, Department of Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for BB-716/A, BB-432B/A, BB-476/A, BB-649A/A and BB-664/A vented, rechargeable nickel cadmium aircraft batteries and cells and harnesses used therein.

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Communications-Electronics Command, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ 07703 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6140

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STANDARDS

FEDERAL

FED-STD-595, Colors Used in Government Procurements

DEPARTMENT OF DEFENSE

MIL-STD-810 Environmental Test Methods and Engineering Guidelines

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.2 Non-Government publications. The following document form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

UL-1642 UL Standard for Safety Lithium Batteries

(Applications for copies should be addressed to Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and applicable specification sheet, the latter shall govern.

3.2 First article. When specified (see 6.6), a sample shall be subjected to first article inspection, in accordance with 4.2.

3.3 Design, materials, and manufacturing. Unless otherwise specified, the design, materials, and manufacturing process selection is the prerogative of the contractor provided all articles submitted to the Government fully meet the operating, interface, ownership and support, and operating environment requirements specified in this document.

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3.3.1 Metals. External metal or component parts of each battery, cell and harness shall show no evidence of cracking, chipping, scaling, corrosion or other deleterious effects when exposed to 1.300 ± 0.025 specific gravity KOH electrolyte solution.

3.3.2 Plastic materials. Plastic materials used in each battery, cell and harness shall show no evidences of cracks, blisters or other deterioration and shall not change more than 2.5 percent in weight or 2.0 percent in any dimension when exposed to 1.300 ± 0.025 specific gravity KOH electrolyte solution.

3.3.3 Elastomer materials. All elastomer materials used in each battery, cell and harness shall show no evidence of cracks, blisters or other deterioration after being exposed to 1.300 ± 0.025 specific gravity KOH electrolyte solution. Materials of pads, gaskets and O-rings shall be selected to provide adequate seal and minimize compression set when tested in accordance with 4.5.3.

3.4 Components. The battery, cells, and harness shall conform to this specification and specification sheets.

3.4.1 Batteries.

3.4.1.1 Battery container and cover. The battery container and cover shall be of alkali-resistant high strength material, free of rough spots, pits, blowholes and other deformations. The container and/or cover shall be completely coated by an epoxy resin to a minimum thickness of 0.015 ± .008 inches to prevent absorption of electrolyte. The dimensions and location of receptacles hold-downs, latches, latch hooks, vent tubes, name and instruction plates, and carrying strap shall conform to the applicable specification sheets. The cover shall be fitted with a hold-down pad having an integral elastomer gasket positioned so as to match the rim of the container body for a gas-tight fit between the cover and the container body. The seal shall prevent the loss of pressure in excess of 2.75 inches of water when tested in accordance with 4.5.8.

3.4.1.2 Latches. The cover shall be secured to the container body by latches mounted on the container body, or by other means specified in the in the applicable specification sheet, and latch hooks mounted on the cover. The latches and hooks shall be mounted on the cover and container body as specified in the applicable specification sheets (see 3.1).

3.4.1.3 Vents. The container body shall be provided with vents to permit gases liberated by the cells of the battery to be purged. These vents shall be located as shown in the applicable specification sheets (see 3.1).

3.4.1.4 Main power receptacle. The battery shall utilize a two-pin receptacle for a main power connector. The receptacle shall be mounted as shown in the applicable specification sheets. The receptacle shall be capable of withstanding a pull out force of 250 ± 25 pounds and a torsional force of 6.0 ± 0.5 foot-pounds.

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3.4.1.5 Intercell connectors. Intercell connectors shall comply with the applicable specification sheets.

3.4.2 Cells. The battery shall contain the number and type of cells in the configurations specified by the applicable specification sheets.

3.4.2.1 Cell mounting. The cells shall be removable but securely mounted in the battery container such that they are not adversely affected by tests performed on the battery to determine its compliance with the requirements of this specification.

3.4.2.2 Cell container. The cell container shall be made of high strength, non-porous, alkali-resistant material. The surfaces of containers shall have a smooth finish, free from pits, blowholes, rough spots, or other deformations.

3.4.2.3 Cell terminals. The size and spacing shall be as specified in the applicable specification sheets. Insulating compounds shall not obstruct the contact surfaces of the terminal.

3.4.2.4 Internal pressure. Each cell shall be assembled and constructed in such a manner that it shall be capable of withstanding an internal pressure of 20 ±1 psig for 15 seconds.

3.4.2.5 Cell vents. Each cell shall be equipped with a vent per applicable specification sheet. The vent plug, when installed to the cell at a torque of 3.5 ± 1.0 in-lb., shall form an airtight seal for the cell when the cell internal pressure is below 2 psig. The vent plug shall be capable of providing automatic relief when the internal cell pressure reaches 6 ± 4 psig. The vent plug shall withstand 7 in-lb of torque with no damage.

3.4.2.6 Potassium carbonate. Cells from the battery shall be tested for potassium carbonate content in the electrolyte. The electrolyte shall not contain more than 60 grams per liter of carbonate.

3.4.3 Harness assembly. The harness assembly shall be capable of withstanding the temperature and exposure of the battery electrolyte without damage (see 6.4).

3.4.3.1 Insulated wires. The harness assembly shall use insulated wires resistant to the battery electrolyte. The insulation shall be free from splinters, blisters and other non-homogeneities visible to the normal eye.

3.4.3.2 Electrical connections. All electrical connections made to receptacle and harness devices (thermistors, thermoswitches ect.) shall be potted with a compound that will not support flame, absorb moisture, or be damaged by battery electrolyte. Contacts shall be so designed that neither the pins nor the sockets will be damaged during mating of counterpart connectors.

3.4.3.3 Thermistors. The harness assembly shall utilize temperature sensor devices that exhibit the resistance-temperature characteristics specified in applicable specification sheet. Potting employed to isolate the thermal sensors from any battery voltage shall conform to 3.6.6.

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3.4.3.4 Sensor receptacle. The battery shall use a hermetic jam nut receptacle to provide connection to battery sensing and control points. The location of the receptacle shall conform to applicable specification sheets (see 3.1).

3.4.3.5 Thermoswitches. Shall conform to the requirements of applicable specification sheet.

3.4.3.6 Device mounting. The devices (thermoswitches and thermistors) shall be mounted on a custom or standard connector strap in such a manner that it shall not become detached during handling. The devices shall be positioned on the connector in such a manner that they shall not interfere with the tool operations required for removal and mounting of cell. If the mounting scheme does not require a connector strap, a separate connector strap shall be provided for each device that is not mounted on a connector.

3.4.3.7 Insulation. All current carrying parts of the harness assembly shall be insulated from the metal parts of the sensor receptacle except for the applicable pins. The insulation resistance between any current carrying part and the sensor receptacle (excluding the applicable pins) shall be above one megohm.

3.4.4 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be employed in intimate contact with each other. This specification terms metals dissimilar when two metal specimens are in contact or otherwise electrically connected to each other in a conductive solution and generate an electric current.

3.4.5 Corrosion prevention. After the battery has been assembled, all exposed metal surfaces of the cells, intercell connectors and associated hardware, and harness shall be coated with a corrosion-preventive compound. The compound shall be uniform, free from grit, abrasives, water, chlorides, or other impurities. The manufacturer is responsible for maintaining carcinogenic free materials. The compound shall be transparent during the protective life of the coating. The compound shall flow, be stable, and uniform. The compound shall contain no precipitates formation or permanent haziness in the solution. The compound shall not show evidence of pitting, etching, or darkening. Care shall be taken not to expose rubber vent sleeves, vent openings or current carrying surfaces to the corrosion preventive film. Coating shall be applied evenly and without voids.

3.4.6 Cleaning. All parts of the battery, cells and harness when assembled shall be cleaned. Corrosive material shall be removed completely. After assembly the battery, cells and harness shall be cleaned thoroughly and shall be free from particles of solder, flux and other foreign material.

3.4.7 Interchangeability. Parts and components of the batteries, cells and harness shall be functionally and dimensionally interchangeable.

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3.4.8 Shipping condition of cells. All cells whether shipped in 19 cell packs or within the batteries shall be delivered complete with electrolyte to proper level and completely discharged, but fully formed. During shipment, a shorting plug or spring shall be connected to the battery's main power receptacle or to the cell's terminals (if shipped separately).

3.4.8.1 Cell electrolyte level. Each cell shall have the electrolyte level adjusted so that after a full charge and a 2 hour stand, the level shall be 1/8 to 1/4 inches above the baffle.

3.5 Color and marking.

3.5.1. Color. The batteries, cells and components shall conform to the requirements in the applicable specification sheets.

3.5.2 Marking. Each battery or cell shall have its identification marking as specified in the following paragraphs and the applicable specification sheets. All markings and identifications shall be permanent and shall be resistant to 1.300 ± 0.025 specific gravity KOH electrolyte. Clear and legible lettering and a method of marking to withstand the operating environment shall be used. A unique serial number is required on each nomenclatured item. The numbers shall be consecutively assigned.

3.5.2.1 Cell Marking.

3.5.2.1.1 Cell terminal marking. Positive terminal markings shall be impressed or embossed with a (+). Negative terminal markings shall be impressed or embossed with a (-). The polarity marking shall be as close as possible to the terminals on the cover of the cell. The positive half of the cell cover shall be permanently marked red. Minimum width of the red band shall be no less than 0.5 inch.

3.5.2.1.2 Cell identification. Each cell shall have the following information printed in black, 1/8-inch lettering:

Battery, Rechargeable (Cell)
**(NSN)*.
Part of BB-xxx/*(AH)/1HR
*(CONTRACT NO.)
*(MANUFACTURER PART NO.)

*FILL IN THE APPLICABLE INFORMATION

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Example label:

Battery, Rechargeable (Cell)
NSN 6140-01-xx-xxxx
Part of BB-xxx
XX AH/1HR
(Contract No.)
(Manufacturer part no.)

Information shall be marked on one of the two largest vertical surfaces of the cell.

3.5.2.2 Battery Marking.

3.5.2.2.1 Battery identification. Each battery shall have its identification marking. The markings shall be printed with white, 1/8 inch lettering and shall have the following information:

Battery, Rechargeable
BB-xxx/ AH/1HR/
*(Contract No.)
*(Date code)
*(Serial No.)
*(Name of manufacturer)
Maximum weight pounds

*Fill in applicable information.

3.5.2.2.2 Caution label. This label shall have black letters on a yellow background. The label shall contain the following lettering size and information:

CAUTION	
DO NOT USE ACID	0.17 ± .030 inch lettering
ELECTROLYTE	
Acid will permanently damage this battery. Avoid using tools, which have been used on acid type storage batteries. Consult technical manual before servicing.	0.09 +. 030 Inch lettering minimum

3.5.2.2.3 Battery polarity marking. The polarity shall be conspicuously and durably marked on the battery container and adjacent to the terminals. Combine with 3.5.2.10, both address the same thing.

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3.5.2.2.4 Battery warning marking. This marking shall be in white print and read as follows:

WARNING

1. Area around battery must be well ventilated and cover removed during charge.

2. Do not allow caustic electrolyte to contact body, clothing, tools, etc. If contact is made wash immediately with water. Flush eyes with water only for 15 minutes and seek medical attention.

3.5.2.2.5 Instruction marking. This marking shall be in white and read as follows:

INSTRUCTION

1. Remove cover and brush off white carbonate deposits with vents in place.

2. Remove shorting plug or spring.

3. Refer to charging instructions in applicable technical manuals.

4. After charge, allow 2 hour stand, remove cell vents and adjust electrolyte using distilled water to 1/4 inches above plates for the cells and 1 inch above plates for the low capacity cell. Replace cell vents and cover.

5. Periodically soak cell vents in distilled water.

3.5.2.2.6 Connector marking. This marking shall be white and centered over the terminal location. The arrows shall point vertically down on each terminal pin.

3.5.2.3 Warning Tags. Where shipping devices are furnished by the contractor to assure proper shorting, disallow cell leakage etc., alkali resistant tags with strong alkali resistant nonmetallic cord shall indicate action or actions to be taken by field personnel to remove and/or replace it.

3.5.2.4 Date code. The date code shall indicate the month and year of manufacture of the battery and cells by means of a four-digit number in which the first two digits indicate the month of the year and the last two digits indicate the year. Months earlier than the tenth shall be indicated by a single digit preceded by "0". When the manufacturer of a battery and cell is completed during the last three working days of the months or the first three working days of the subsequent month, the manufacturer is permitted to use either month as the coded month of

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manufacture. The date code shall be located on the cover of the cell. The numbers shall be 1/8 inches in height and black in color. For the low capacity cell the numbers shall be white in color.

3.6 Performance.

3.6.1 Capacity. When discharge of battery and cells is specified in this specification, the battery and cells shall be discharged at the current rate specified in the applicable specification sheet to the specified cut-off voltage for the test listed. The battery and cell capacity shall be equal to the specified discharge current (amperes) multiplied by the discharge time above the specified cut-off voltage (hours). The battery and cells shall provide the capacity specified in the applicable specification sheets.
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3.6.2 Environmental Requirements. The battery and cells when subjected to the environmental tests in the verification tables shall show none of the following:

a. Dimensional distortion beyond specified limits or cracking of case or cover of either cell or battery.

b. Radical current or voltage fluctuations during any test.

c. Mechanical failure of any part.

d. Electrolyte leakage or any spilling of electrolyte at any time during test. Venting of gasses containing entrained electrolyte is permitted during charge.

e. Breakdown of insulation, stripping of metal plating from any component part, corrosion of metal parts, or loosening of protective coating from the battery container or cover.

f. Deterioration of battery/cell identification markings.

3.6.2.1 Harness testing. The harness assembly when subjected to the temperature shock test of 4.5.23 or alkali resistance test of 4.5.21 shall show none of the following:

a. Breakdown of insulation, stripping of metal plating from connectors, corrosion of metal parts or cracking of potting compound.

b. Failure of thermistors to meet the requirements of 3.4.3.3.

c. Failure of thermoswitches to meet the requirements of 3.4.3.5.

d. Mechanical failure of components.

e. Dimensional distortion beyond specified limits.

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3.6.3 Electrolyte leakage. The battery and cells shall show no evidence of electrolyte leakage during the performance of any of the tests specified in Section 4. Venting of gasses containing entrained electrolyte is permitted during charge.

3.6.4 Operating position. The battery shall be capable of operating in an inverted position without loss of any electrolyte from the cells. The cells shall be capable of operating in an inverted position without loss of electrolyte.

3.6.5 Insulation. All current carrying parts shall be insulated from the battery container. The insulation resistance between any current carrying part and the battery container shall be above one greater than one megohm.

3.6.6 Potting. Compounds used for potting shall not support flame, absorb moisture, or be damaged by battery electrolyte. When mixed in the ration specified by the manufacturer, the product shall cure at a temperature less than 130 degrees Celsius (°C). The components, individually or when mixed, shall have no adverse effect on the health of personnel when used for its intended purpose. The packaged components shall be free from all foreign substances, such as grit, dirt, oil, and water. The components shall show no gel particles, ingredient lumps, or agglomerates. Asbestos shall not be used in the product formulation. The components shall be free of nonreactive, volatile solvents.

3.6.7 Wiring and electrical connections. Wiring shall reveal no readily visible cracks in the insulation when visually examined. Specimens shall reveal no delaminating of the insulation, no exudation of the insulation throughout the braid in constructions incorporating braids, and no visible defects in any of the wire component parts (slight discoloration after aging is considered normal and acceptable). All electrical connections shall be potted per 3.6.6.

4. VERIFICATION.

4.1. Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

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Table I. First article inspection of components and materials.

Test	Part	Requirement Paragraph	Test paragraph
Corrosion resistance	Metal parts 2/	3.3.1	4.5.1
Alkali absorption of plastic materials	Plastic parts	3.3.2	4.5.2
Alkali resistance of elastomer materials	Elastomers	3.3.3	4.5.3
Temperature shock of harness 1/	Harness	3.6.2.1	4.5.23
Alkali resistance of harness 1/	Harness	3.3.1, 3.3.2, 3.6.6, 3.6.2.1	4.5.21
Insulation resistance of harness 1/	Harness	3.4.3.7	4.5.22
Venting test	Vent plug	3.4.2.5	4.5.4.1
Torque test	Vent plug	3.4.2.5	4.5.4.2

NOTES: 1/ Test conducted except for cell only procurements.
2/ Container, cover and latches, external cell terminal hardware, intercell connectors and hardware, etc.

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TABLE II. First Article Verification of batteries. /Components

Test	Sample or Cell Group Number 1/	Requirement Paragraph	Test Paragraph
Visual and mechanical	1-4	3.4, 3.5, 3.6.3	4.5.5
Dimension and weight	1-4	3.4	4.5.7
Insulation resistance 2/	1-4	3.6.5	4.5.6
Receptacle mounting 2/	1-4	3.4.1.4	4.5.20
Container body to container cover 2/	1-4	3.4.1.1	4.5.8
Internal pressure 3/	1-4	3.4, 3.4.2.4	4.5.19
Capacity (initial)	1-4	3.6.l, 3.6.3, 3.4.8.1	4.5.10.1
Potassium carbonate 3/	1-4	3.4.2.6	4.5.9
High rate discharge, low temperature	2,4	3.6.1, 3.6.2	4.5.17
High rate discharge operating position	1,3	3.6.1, 3.6.4	4.5.18
Cycle life	1,2	3.6.1, 3.6.3	4.5.11
Mechanical shock	3,4	3.6.2	4.5.12
Vibration	3,4	3.6.1, 3.6.2	4.5.13
Temperature shock	3,4	3.6.1, 3.6.2	4.5.14
Altitude	3,4	3.6.1, 3.6.2	4.5.15
Capacity	3-4	3.6.1, 3.6.3	4.5.10
Humidity and charge retention	3,4	3.6.1, 3.6.2	4.5.16
Capacity	3-4	3.6.1, 3.6.3	4.5.10

NOTES: 1/ Cell group (see 4.4.2.1)
2/ Not applicable to cell procurements
3/ For battery or battery plus component procurements test four cells and one low capacity cell per battery sample. For separate cell procurement test four cells per cell group.

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TABLE III. Identification defects.

Number	Description	Method of inspection
1	Electrical contact surfaces obstructed by Insulation compounds.	Visual
2	Pitting or blow holes on the external calls container.	Visual
3	Electrolyte leakage and quantity of electrolyte.	Visual
4	Location and polarity of terminals not as specified.	Visual
5	Terminal, identification, caution and warning markings not as specified.	Visual
6	Terminal seals missing or defective.	Visual
7	Corrosion.	Visual
8	Particles of foreign material.	Visual
9	Insulators or insulation missing or damaged.	Visual
10	Welds containing blowholes, cracks, or slag inclusions.	Visual
11	Burrs on battery container, cover or interconnecting hardware.	Visual
12	Improper color on outside of container and cover.	Visual
13	Thermal switches open at room temperature.	Meter
14	Incorrect thermistor values.	Meter
15	Date codes missing or incorrect or more than one date code on a battery.	Visual
16	Carrying strap on cover missing, twisted or torn. Carrying strap stitches are loose or broken, strap cut edges are not sealed to prevent unraveling.	Visual
17	Connector mounted incorrectly (key not oriented at 12 o'clock position).	Visual

4.2 First article inspection. This inspection shall be performed by the contractor unless otherwise specified in the contract. The inspection shall consist of the following:

a. Inspection and testing of batteries specified in Table II and shall be performed in the order indicated.

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b. Inspection and testing of three samples of each applicable battery components and materials treated and processed as they would be for fabrication of finished battery in accordance with Table I. All samples shall be marked properly with identifying information.

4.2.1 First article samples. Unless otherwise specified (see 6.2), the contractor shall furnish four first article samples selected at random of each battery type on order.

4.2.1.1 Separate cell procurements. For separate cell procurements, the contractor shall furnish 76 first article samples selected at random. The cells shall be organized into four equal number cell groups (19 cells each) for inspection purposes.

4.2.1.2 Separate cell procurements. For separate procurements of cells and batteries under the same contract, the battery first article tests will also qualify the cell first article test requirements.

4.2.1.3 Battery plus component procurements. For procurements of batteries plus components (cells, low capacity cells and/or harnesses) from the same manufacturer, the batteries shall be subjected to first article sampling per 4.2.1 and the harness shall be subjected to first article sampling per 4.2.1.4.

4.2.1.4 Separate harness procurements. For separate harness procurements, the contractor shall furnish three first article samples selected at random.

4.2.2 Definition of failure. First article sample failure is defined as follows:

a. Failure to deliver the electrical performance specified in 3.6.1.

b. Distortion, cracking or breakage which prevents the battery or any of its components from mating with a corresponding part or maintaining its electrical leak proof requirements.

c. The batteries, harnesses, or cells shall be considered unacceptable if any of its first article samples fail to pass the inspections of Table I or II.

4.3 Quality conformance inspection.

4.3.1 Quality conformance inspection before packaging. Inspection of product delivery shall consist of groups A, B, and C inspections. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government shall review and evaluate the contractor's inspection records. In addition, the government at its discretion may perform all or part of the specified inspection, to verify the contractor's compliance with specified requirements (see 6.5). Test equipment for government verification inspection shall be made available by the contractor at no cost.

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4.3.2 Inspection lot definition. An inspection lot shall be defined as the quantity of batteries, cells and harnesses or any one type produced at any one place of manufacture on any one contract, submitted at one time to quality conformance inspection. Batteries or cells with no more than two consecutive date codes shall appear in a particular inspection lot. A specific date code shall not be found in any more than one inspection lot.

4.3.2.1 Separate cell procurements. For separate cell procurements, a cell inspection lot shall be defined as the number of cells produced divided by 19 to determine lot size for inspections. Each group of 19 cells shall be considered as a battery for inspection purposes.

4.3.2.2 Separate harness procurements. For separate harness procurements, a harness shall be considered as a battery for inspection purposes.

4.3.2.3 Battery plus component procurements. For procurements of batteries plus components parts (cells, and harness assemblies) from the same manufacturer, only the batteries shall be subjected to Group A, B and C inspections. The batteries shall qualify all.

4.3.3 Group A inspection. Each unit of inspection lot shall be subjected to and meet the requirements of Table IV. Defective units shall be individually rejected. Group A inspection shall be performed in any order which is satisfactory to the government.

4.3.4 Group B inspection. This inspection, including sampling, shall conform to Table V. Inspection shall be performed on inspection lots that have passed Group A inspection. Samples for subgroup II are to be selected from units that have passed subgroup I inspection, and the remainder of subgroup I samples may be furnished under contract. No shipment shall be made until Group B inspection is completed and approved by the government inspector.

4.3.5 Group C inspection. Group C inspection shall be performed in accordance with Table VI on two (2) units that have been randomly selected from the lot. One unit shall be assigned to subgroup I, one to subgroup II. The samples may be selected before full marking and Group A inspection has been applied to the lot. However, the test items shall be fully completed prior to the start of Group C testing. Shipment of batteries or cells shall not be made until subgroups I have met the required inspections and 100 cycles of the required number of cycles of subgroups II have been successfully completed. Inspection shall be performed in the order shown.

TABLE IV Group A Inspection.

Test	Requirement, Paragraph	Test Paragraph
Visual and Mechanical	3.4, 3.5, 3.6.3	4.5.5
Insulation Resistance 1/	3.6.5	4.5.22

NOTES: 1 /, Not Applicable to cell procurements
Sample size shall be determined by TABLE VII

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TABLE V. Group B Inspection.

Test	Requirement Paragraph	Inspection Paragraph
Dimension and weight	3.1	4.5.7
Container body to Container Cover	3.4.1.1	4.5.8
Internal pressure	3.4, 3.4.2.4	4.5.19
Vent Plug Test	3.4.2.5	4.5.4
Capacity (Initial)	3.6.1, 3.6.3, 3.4.9	4.5.10.1
Subgroup II		
High Rate Discharge, Low Temperature	3.6.1, 3.6.2	4.5.17
High Rate Discharge Operating Position	3.6.1, 3.6.4	4.5.18

NOTES: 1/ Not applicable to cell procurements.
2/ For battery or battery plus components, test four cells and low capacity cell per.battery sample. For separate cell procurements, test four cells per cell group

TABLE VI. Group C Inspection.

Test	Requirement Paragraph	Test Paragraph	Subgroup I	Subgroup II
Potassium Carbonate 1/	3.4.2.6	4.5.9	X	X
Capacity	3.6.1, 3.6.3	4.5.10	X	X
Humidity and Charge Retention	3.6.1, 3.6.2	4.5.16	X	
Cycle Life, 200 cycles	3.6.1, 3.6.3, 3.6.8	4.5.11		X
Temperature Shock 2/	3.6.1, 3.6.2, 3.4.3, 3.6.2.1	4.5.14	X	
Vibration 3/	3.6.1, 3.6.2	4.5.13	X	
Alkali Resistance of harness	3.3.1, 3.3.2, 3.6.6, 3.6.2.1	4.5.21		X

NOTES: 1/ For battery or battery plus component procurements, test 4 cells and 1 low capacity cell per battery sample. For separate cell procurement test 4 cells per cell group (See 4.5.2.1)

2/ For separate harness assembly procurements (See 4.5.23) for test method.

3/ Not applicable for cell procurements.

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TABLE VII. Sampling Plan.

Table	Group	Lot Size	Sample Size
Table I	Battery Components and Materials	-	3
Table II	First Article Batteries/Cells	-	4
Table IV	Group “A”	-	100%
Table V	Group “B:	1-20 21-50 51-100	6 8 10
Table VI	Group “C”	-	2

4.4 Inspection conditions and equipment.

4.4.1 Inspection and test conditions.

4.4..1.1 Normal temperature. Unless otherwise specified, all measurements and tests shall be made at 73 ± 9°F (23 ± 5^0C) at ambient atmospheric pressure and relative humidity with a minimum storage of 30 minutes between charge and discharge.

4.4.1.2 Charging.

4.4.1.2.1 Constant current method. The batteries shall be charged constant current at the C amperes to 1.55 ± .0l volts per cell then 1.1 amperes for another 3 hours. Unless specified otherwise, the electrolyte shall be adjusted with distilled water or deionized to the levels specified by 3.4.8.1 and 3.4.8.2 two hours after charge termination.

4.4.1.2.2 Fast charge method. The batteries shall be charged at 1.5 C amperes to 1.55 ± .0l0 volts per cell, then continue for another minute at the same rate.

4.4.1.3 Reconditioning. If a sample fails to meet the specified capacity or voltage requirement, it shall be subjected to reconditioning prior to repeating the capacity or voltage test (rerun). Reconditioning shall consist of deep discharge on an individual cell basis for batteries or individual cells. Each cell shall be discharged to zero volts within 24 or less hours, followed by a full charge (see 4.4.1.2.1) and adjustment of electrolyte level. Should the sample not meet the capacity or voltage requirement upon rerun, the repeated test shall be considered a failure. If the sample passes rerun, the specified test schedule shall be resumed.

4.4.1.4 Cell discharge. For all discharges conducted on the battery in the tests specified in this specification and applicable specification sheets, the voltage and length of discharge prior to the cut-off voltage specified of the cell shall be monitored.

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4.4.2 Test equipment and inspection facilities. Test equipment and inspection facilities shall be of sufficient accuracy, quality, and quantity to permit performance of required inspection. Test equipment for Government verification inspection shall be made available by the supplier.

4.4.3 Instrument Accuracy.

4.4.3.1 Electrical indicating instruments. All voltmeters and ammeters shall be accurate within 0.5 percent of the full scale reading. The range shall be such that the readings are taken on the upper half of the scale. Timers shall be accurate within 0.5 percent. The sensitivity of voltmeter shall be at least 1000 ohms per volt.

4.4.3.2 Resistance tolerances. In all tests involving discharge through a resistance, the total circuit resistance shall be accurate within 0.5 percent.

4.5 Examination and test methods.

4.5.1 Corrosion resistance. Samples of all metal parts of the cells and/or battery shall comply with 3.3.1 after being subjected to this test. Additionally, a four-by six-inch panel shall be furnished for this test, coated as it would be supplied on the battery and containing sample identification markings. Soak one-half of each specimen along its horizontal length with a solution of 1.300 ± 0.025 specific gravity KOH electrolyte and allow to stand for 7 days at 65.5° ± 1.1°C (150° ± 2°F). Specimens shall then be rinsed, dried, and observed for compliance with 3.3.1.

4.5.2 Alkali absorption of plastic material. Three specimens each two by three inches, shall be cut from each different plastic material submitted by the contractor. The cut edges may be sealed with the same resin used in fabrication of the case. Plastic materials normally containing identification markings shall be supplied with sample markings for this test. After being measured with a micrometer and weighed in a dry condition (dried in a temperature chamber for 20.5 hours at 32.2° ± 1.1°C (90° ± 2°F) on a chemical balance, each specimen shall be immersed in a covered vessel containing 150 cc of 1.300 ± 0.025 specific gravity KOH electrolyte. This vessel shall be held for 7 days in a temperature chamber at 65.5° ± 1.1°C (150° ± 2°F). At the end of the heating period, the specimens shall be rinsed in water and dried on the surface. The identification marking sample shall be observed for the requirements of 3.5.2. All samples shall then be inspected for evidence of cracks or blisters and shall be measured and weighed, and the percentage change in dimensions and weight shall be calculated to determine conformance to the requirements specified in 3.3.2.

4.5.3 Alkali resistance of elastomer material. Three samples each of pads, gaskets, O-rings, etc., made of elastomer material used in the battery or cell shall be immersed for 7 days in 1.300 ± 0.025 specific gravity KOH electrolyte in a temperature chamber at 65.5° ± 1.1°C (150° ± 2°F). They shall then be rinsed with water and allowed to dry. They shall meet the requirements of 3.3.3.

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4.5.4 Vent plug test.

4.5.4.1 Venting tests. The vent of each cell being tested shall be installed in an adapter through which controllable air pressure may be applied to the vent. The air pressure shall be raised from 0 to 10 psig and returned to 0 psig in a steady and uniform manner over a 2.0 minute period. The test shall be performed at ambient air temperature and repeated at -30°C (-22°F) except that the vent shall be conditioned at -30°C (-22°F) in air for 48 hours prior to testing. The cell vents shall conform to 3.4.2.5.

4.5.4.2 Torque test. The vent plug of each cell being tested shall be closed to a torque of 7 in-lb and then opened. This closing and opening shall be repeated a total of five times and the vent plug shall meet the requirements of 3.4.2.5. The vent plugs may be screwed into a threaded metal adapter, rather than a cell, to conduct this test.

4.5.5 Visual and mechanical. Samples shall be examined to verify that the basic materials, component parts, finish, and marking in accordance with all requirements of 3.4, 3.5, 3.6.3 and Table III.

4.5.6 Insulation resistance. Insulation resistance of a discharged battery shall be measured between the battery terminal and the exposed metal of the latch, and between sensor receptacle contact pins E or P and the exposed metal of the latch. Insulation resistance measurements shall be made on an apparatus suitable for the characteristics of the component to be measured such as a megohm bridge, megohm-meter, insulation-resistance test set, or other suitable apparatus. Unless otherwise specified, the direct potential applied to the specimen shall be 500 volts ± 10%. Each measurement shall be taken every 2 minutes, shall last 20 seconds, and shall meet the requirements of 3.6.5.

4.5.7 Dimension and weight. The battery shall be examined to verify compliance with the requirements of 3.4.

4.5.8 Container body to container cover seal. The battery, with the cover secured (latches applied normally), shall have a manometer attached to one of the battery vents and air pressure, equivalent to 55 inches of water, applied through the other battery vent and sealed at that level for two minutes. The manometer shall be read at the instant of attaining and sealing in the required pressure and again at two minutes after the sealing. The battery shall meet the requirements of 3.4.1.1.

4.5.9 Potassium carbonate. Conduct the capacity discharge test the required number of additional times to provide a total of three discharges. Following the third discharge, the battery shall stand on open circuit for one hour, after which the battery shall be charged. Following a one-hour stand, electrolyte from five cells shall be tested .

4.5.10 Capacity. The battery shall be charged per 4.4.1.2.1 and discharged at the rate specified by the applicable specification sheet to the cut-off voltage specified. The battery shall meet the requirements of 3.6.1 and 3.6.3.

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4.5.10.1 Initial capacity. The battery shall be charged per 4.4.1.2. and shall reach an average voltage level of 1.55 ± .0l volts/cell within 1 hour and 40 minutes of the start of charging in order to demonstrate conformance with the requirement for full formation of the cells. After 2 hours rest, the electrolyte level of the cells and low capacity cell shall meet the requirements of 3.4.8.1 and 3.4.8.2. The battery shall be discharged at the rate specified by the applicable specification sheet to the cut-off voltage specified. The battery shall meet the requirements of 3.6.1 and 3.6.3.

4.5.11 Cycle life. The battery shall be tested as follows:

a. Cycle 1. Charge per 4.4.1.2.1. Rest for one hour at open circuit. Then discharge at the normal rate specified by the applicable specification sheet to the cut-off-voltage specified.

b. Cycle 2. Charge per 4.4.1.2.1. Rest for one hour at open circuit. Then discharge at the high rate specified by the applicable specification sheet for 2.0 minutes.

c. Cycles 3-98. Charge per 4.4.1.2.2 and rest for 1 hour at open circuit and discharge at the high rate specified by the applicable specification sheet for 2.0 minutes.

d. Cycle 99. Charge per 4.4.1.2.2 and rest for one hour at open circuit and discharge at the high rate specified by the applicable specification sheet to the cut-off voltage specified.

e. Cycle 100. Repeat step a.

f. Cycle 101. Repeat step b.

g. Cycle 102 -198. Repeat step c.

h. Cycle 199. Repeat step d.

i. Cycle 200. Repeat step a.

NOTE: Water may be added on the 100th cycle. The amount of water added shall be recorded. Reconditioning per 4.4.1.3 is not allowed during the cycle life test.

4.5.11.1 Failure. If any of the following conditions occur during the cycle test, the battery shall be judged to have failed:

a. The voltage during discharge is below the cut-off voltage specified in the applicable specification sheet.

b. The capacity of either the l00th or 200th cycle is less than the capacity specified on the applicable specification sheet.

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c. The battery or cell when inspected per 4.5.5 fail to meet the requirements of 3.4, 3.5, and 3.6.3.

4.5.12 Mechanical shock. The battery shall be charged per 4.4.1.2.1 and then subjected to a mechanical shock test per procedure I, method 516 of MIL-STD-810 using figure 516.4. The test sample shall be mounted and shocked in the normal upright position. During test the open circuit voltage shall be monitored. The battery shall conform to the requirements of 3.6.2

4.5.13 Vibration. The battery shall be charged per 4.4.1.2.1 and then subjected to a Vibration Test in accordance with the vibration test method of UL-1642, using a low temperature of -70°F (-57°C) and a high temperature of 160°F (71°C). The battery shall be discharged at the rate specified by the applicable specification sheet to the cut-off voltage specified and shall meet the requirements of 3.6.1 and 3.6.2. The open circuit voltage shall be monitored during test.

4.5.14 Temperature shock. The battery shall be charged per 4.4.1.2.1 and subjected to the temperature shock test using a low temperature of -70°F (-57°C) and a high temperature of 160°F (71°C). Diurnal cycling is not required. Upon completion the battery shall be stabilized at normal conditions per 4.4.1.1 and discharged at the rate specified by the applicable specification sheet to the specified cut-off voltage and shall meet the requirements of 3.6.1 and 3.6.2. The open circuit voltage shall be monitored during test. The components (e.g. thermistors, thermoswitches and etc.) of the harness in the battery shall be monitored during the Thermal Shock Test. The components shall meet the performance requirements specified in 3.4.3.

4.5.15 Altitude. The battery shall be tested as follows:

a. Charge per 4.4.1.2.1

b. Place battery with cover in place (on open circuit) into environmental chamber and within 15 minutes; adjust chamber pressure to simulate pressure at 60,000 feet above sea level.

c. Discharge at the rate specified by the applicable specification sheet to the specified cut-off voltage. The battery shall meet the requirements of 3.6.1.

d. Charge per 4.4.1.2.2 under the conditions of step b.

e. Discharge per step c.

f. After discharge return the chamber to normal conditions specified in 4.4.1.1 within 5 minutes.

g. The battery shall be inspected for defects per 4.5.5 and shall meet the requirements of 3.6.2.

4.5.16 Humidity and charge retention. The battery shall be charged per 4.4.1.2.1 and then subjected to the humidity test per procedure II, method 507 of MIL-STD-810. The battery

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case air vents shall be positioned where the axis of the vents is parallel to the chamber airflow. Upon completion, accumulated water may be removed from the battery. The battery shall be discharged at the rate specified by the applicable specification sheet to the specified cut-off voltage and shall meet the requirements of 3.6.1 and 3.6.2.

4.5.17 High rate and low temperature discharge. The battery shall be charged per 4.4.1.2.1 and stored at -22+2°F (-30+ 1.1°C) for 16 hours. The battery shall be discharged at the rate specified by the applicable specification sheet to the specified cut-off voltage and shall meet the requirements of 3.6.1 and 3.6.2.

4.5.18 High rate discharge and operating position. The battery shall be charged per 4.4.1.2.1 and discharged at the rate specified by the applicable specification sheet to the specified cut-off voltage. The battery shall be in an inverted position for the first 2.5 minutes of discharge and shall meet the requirements of 3.6.1 and 3.6.4.

4.5.19 Internal pressure test. Adequately restrained cells shall have dry nitrogen or air at an internal pressure of 20 ±1 psig at ambient temperature applied to them through an airtight adapter to their venting aperture. The source of pressure shall be shut off and the pressure maintained in the cells for a minimum of 15 seconds. The cells shall meet the requirements of 3.4.2.4.

4.5.20 Receptacle mounting test. A special grooved shaft similar to the clamping shaft of the battery quick disconnect plug, shall be so inserted in the receptacle socket that the socket pins are secured in the steel shaft. A tensional force of 250 ± 25 pounds shall be exerted against the socket pins. Following this test, the steel shaft shall be subjected to torsional force of 6 ± 0.5 foot-pounds. The receptacle mounting shall meet the requirements of 3.4.1.4.

4.5.21 Alkali resistance of harness. The harness assembly, with the exception of the J2 connector, shall be immersed in a covered vessel containing 1.300 ± 0.025 specific gravity KOH electrolyte. The connector shall be on the outside of the covered vessel containing the solution and protected from contact with either the solution or vapors from the solution. The vessel shall be stored in a chamber for 7 days at a constant temperature of 150 ± 5°F (64.5 ± 3.3°C). At the end of storage, the harness shall be rinsed with water and dried. The harness assembly shall be inspected for cracks or blisters, electrical continuity and isolation (where applicable) and shall meet the requirements of 3.3.1, 3.3.2, 3.6.6 and 3.6.2.1.

4.5.22 Insulation resistance of harness. The insulation resistance of the harness assembly shall be measured between the metal area of the sensor receptacle (exclude applicable pins) and any current carrying part consistent with the insulation resistance test, for one minute and shall meet the requirement of 3.4.3.7.

4.5.23 Harness temperature shock. The harness assembly shall be subject to the Thermal Shock Test consistent with the battery temperature shock test. Each thermistor and thermoswitch shall be monitored. The harness assembly shall meet the requirements of 3.6.2.1.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature, which may be helpful, but is not mandatory)

6.1 Intended use. The batteries, cells and harness covered by this specification are used in US Army aircraft. They are military unique because they must be able to operate reliably and satisfactorily on US Army aircraft, subject to severe flight maneuvers, environments and battlefield conditions.

6.2 Acquisition requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced.

c. Packaging requirements (see 5.1)

d. First article inspection (see 3.2), sample quantities (if sample quantities differ from 4.2.1 requirements) and Disposition Instructions for Groups A, B and C test samples.

e. When certification of parts, materials, or components is not required (see 3.3).

6.3 Definitions.

6.3.1 Fully charged batteries. Batteries should be considered fully charged when charged in accordance with 4.4.1.2.1.

6.3.2 Discharged batteries. Batteries should be considered discharged when a specified discharge rate was conducted until a specified final voltage was reached. For shipping purposes only, the battery should be discharged to zero volts per cell.

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6.3.3 End voltage. The end voltage of the battery is the specified end voltage terminating a specified discharge rate.

6.3.4 Cycle. A cycle is a combination of a charge and discharge.

6.3.5 Normal upright position. Perpendicular to the plane of the horizon.

6.3.6 Fully formed. Batteries and cells should be considered fully formed if they have been charged to the point where all available cadmium oxide has been converted to cadmium metal and both hydrogen and oxygen gases were evolved at close to stoichiometric ratios.

6.4 Harness assembly. Referenced tables and figures, refer to applicable specification sheets.

6.5 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government may be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.6 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of four units. The contracting officer should include specific instructions in the acquisition documents regarding arrangements for examinations and approval of the first article test report.

6.7 Subject term (key word) listing.

Battery
Cells
Harnesses
Nickel-Cadmium

Custodian
Army - CR

Preparing Activity:
Army - CR

(Project 6140-A917)

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-49450B	**2. DOCUMENT DATE (YYMMDD)** 00/03/10

3. DOCUMENT TITLE
BATTERY, RECHARGEABLE, NICKEL-CADMIUM, VENTED, AIRCRAFT

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME US Army Communications-Electronics Command	b. TELEPHONE (Include Area Code) (1) Commercial (732) 532-9139 (2) AUTOVON 992-9139
c. ADDRESS (Include Zip Code) ATTN: AMSEL-LC-LEO-E-EP Fort Monmouth, NJ 07703-5023	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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